

Morphic Studio

Document 02B — Product Requirements Document (PRD)

Functional Requirements

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1. Purpose

This document defines the functional requirements of Morphic Studio.

A functional requirement describes a capability the platform **must** provide. These requirements define the expected behavior of the system from the user's perspective and serve as the primary implementation checklist during development.

2. Functional Requirement Principles

Every requirement should follow these principles:

- Be user-centered.
 - Be testable.
 - Be modular.
 - Be reversible where possible.
 - Preserve project consistency.
 - Keep the user in control.
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3. User Account Requirements

FR-001 — User Registration

The platform shall allow users to create an account using supported authentication providers.

The system shall:

- Create a unique User ID.
 - Create a default workspace.
 - Initialize user preferences.
 - Create a personal Asset Library.
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FR-002 — Authentication

The platform shall securely authenticate users and protect access to their projects.

Users shall remain signed in until they explicitly sign out or their session expires.

4. Project Management Requirements

FR-010 — Create Project

Users shall be able to create a new project by providing:

- Project name
- Description (optional)
- Project type (Comic, Animation, Motion Comic, Hybrid)
- Preferred language
- Target audience (optional)

When a project is created, the platform shall automatically initialize:

- Project ID
 - Project Brain
 - Story Bible
 - Project DNA
 - Empty Asset Library
 - Version History
 - Creative Timeline
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FR-011 — Project Dashboard

Each project shall display:

- Current progress
- Story status
- Asset count
- Character count

- AI tasks
 - Recent activity
 - Export status
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5. Project Brain Requirements

Each project shall contain exactly one Project Brain.

The Project Brain is responsible for maintaining:

- Current production state
- Story progress
- Creative intent
- User decisions
- Active workflows
- Connected assets
- AI context

The Project Brain shall synchronize with Morphic Core after significant project changes.

6. Project DNA Requirements

Each project shall contain a Project DNA profile.

The Project DNA shall define the creative identity of the project.

It shall include:

Visual DNA

- Art style
- Character proportions
- Lighting
- Color palette
- Camera language
- Composition rules

Story DNA

- Genre
- Themes
- Narrative tone

- Pacing
- Target audience
- World rules

Audio DNA

- Voice style
- Music direction
- Sound design
- Ambient preferences

Production DNA

- Preferred output formats
- Export quality
- Animation style
- Motion comic preferences

AI DNA

- Preferred AI models
- Creativity level
- Consistency settings
- Generation policies
- Safety preferences

Project DNA shall be editable at any time.

Changes shall apply only after user confirmation.

7. Story Bible Requirements

Every project shall include one Story Bible.

The Story Bible shall maintain:

- Characters
- Locations
- Timeline
- Relationships
- World rules
- Organizations
- Objects
- Events
- Lore

Whenever the user changes story information, the Story Bible shall update automatically after confirmation.

8. Character Management Requirements

Each character shall receive a unique Character ID.

Character records shall include:

- Name
- Nickname
- Age
- Gender
- Appearance
- Clothing
- Expressions
- Personality
- Voice
- Relationships
- Goals
- Backstory
- Current status

The platform shall prevent accidental duplication of existing characters.

If two characters intentionally share the same name, their Character IDs shall remain distinct, ensuring internal consistency.

9. Asset Library Requirements

The Asset Library shall store reusable creative assets.

Supported categories include:

- Characters
- Outfits
- Expressions
- Backgrounds
- Props
- Voices
- Music
- Sound Effects
- Animations
- Logos

- Icons

Assets shall be searchable, taggable, versioned, and reusable across the same project.

10. AI Workflow Requirements

Whenever a user requests AI generation, Morpheic Core shall execute the following sequence:

1. Read Project Brain.
2. Read Story Bible.
3. Read Project DNA.
4. Search the Asset Library for reusable content.
5. Identify missing information.
6. Request clarification if needed.
7. Generate only the missing assets.
8. Save approved outputs.
9. Update project memory.

AI modules shall never bypass this workflow unless explicitly directed by the user.

11. Creative Timeline Requirements

Every significant creative action shall create a checkpoint.

Examples include:

- Story imported
- Story Bible updated
- Character approved
- New asset generated
- Comic page finalized
- Animation exported

Users shall be able to:

- View timeline history.
- Restore earlier checkpoints.
- Compare checkpoints.
- Create alternate project branches.

No checkpoint shall permanently overwrite previous work.

12. Export Requirements

The platform shall support exporting projects in formats appropriate to the completed work.

Export settings shall remember:

- Resolution
- Aspect ratio
- Compression
- Watermark preferences
- File naming conventions

Future export formats can be added without redesigning the export system.

13. Error Handling Requirements

When required information is missing or conflicting, the system shall:

- Detect the issue.
- Explain the conflict.
- Recommend possible solutions.
- Allow the user to choose how to proceed.

The platform shall avoid silently making irreversible creative decisions.

14. Functional Success Criteria

The platform meets its functional requirements when a user can:

- Create an account.
 - Create a project.
 - Build a Story Bible.
 - Establish Project DNA.
 - Generate consistent characters.
 - Reuse assets.
 - Produce comics.
 - Track project history.
 - Restore earlier versions.
 - Export finished work.
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